Alexander Junge

I combine my passion for data science with a strong analytical mindset.

Key competencies

- Computational biology
- Development of reproducible workflows
- Supervised machine learning
- Leading an internat. student organization
- Data cleaning and analysis
- Dimensionality reduction
- Cluster analysis
- Proactive and detail oriented

Education

03/2014–02/2017 (estimated) **PhD student in theoretical and applied RNA Bioinformatics**, *University of Copenhagen*, Frederiksberg, Denmark, Supervisor: Professor Jan Gorodkin, PhD; Co-supervisors: Jack Cowland, PhD and Jakob Hull Havgaard, PhD.
My project combined the computational analysis of RNA structure and function with clinical applications to leukemia, in cooperation with the Copenhagen University Hospital. I successfully combined work on my PhD project with presenting my research at international conferences, the completion of scientific manuscripts, and the organization of local and international scientific events.

10/2011–02/2014 **Master's studies of Bioinformatics**, *Saarland University*, Saarbrücken, Germany.
Full Master's degree program (120 ECTS credits), Final GPA: 1.1 (A)

03/2013–08/2013 **Master's thesis stay**, *University of Southern Denmark*, Odense, Denmark, Research group on *Computational Biology* headed by Associate Prof. Dr. Jan Baumbach.
Developed a runtime- and memory-efficient clustering algorithm that handles missing distance information. I also implemented the algorithm in Java.

08–12/2011 **Erasmus exchange semester abroad**, *Linköping University*, Linköping, Sweden.

10/2008–08/2011 **Bachelor's studies of Bioinformatics (Computational Molecular Biology)**, *Saarland University*, Saarbrücken, Germany.
Full Bachelor's degree program (180 ECTS credits), Final GPA: 1.4 (A-)

06/2007 **Abitur (general qualification for university entrance; comparable to A-levels)**, *Ludwig-Wilhelm-Gymnasium*, Rastatt, Germany.

Experience

01/2015 and 01/2016 **Lecturer**, *Lecture 'Structural Bioinformatics'*, Bioinformatics Master's program, University of Copenhagen, Denmark.
Taught RNA structure prediction and led corresponding programming labs in Python.

03/2012–09/2013 **Student research assistant**, *Research group on Computational Systems Biology*, Headed by Assoc. Prof. Dr. Jan Baumbach at the Max Planck Institute for Informatics, Saarbrücken, Germany.
Co-developed KeyPathwayMiner - a Java application to identify disease-related pathways by integrating diverse clinical data sets.

04/2010–03/2011 **Teaching assistant**, *Lecture 'Programming 2'*, Taught by Prof. Dr. Sebastian Hack at Saarland University, Saarbrücken, Germany.
Introduced undergraduates to the basics of object-oriented programming, primarily using Java.

alexander.junge@gmail.com • *alexanderjunge.net* • *alexanderjunge*
@JungeAlexander

Technical Experience

Proficient With

Languages	Python, R
Technologies	Git, LaTeX, Linux, Nextflow, PBS/Torque, PyData stack, Vim, Windows

Have Experience With

Languages	Bash, C, C++, CSS, Java, JavaScript, HTML, Matlab, Perl, PHP
Technologies	Bootstrap, Django, Eclipse, jQuery, Make, MySQL, Slurm, SQLite

Languages

German	Native language	Danish	Intermediate knowledge
English	Full professional proficiency	French	Basic knowledge

Extracurricular Activities

01/2015–present **International Society for Computational Biology (ISCB) Student Council**, *Vice Chair & Chair of the Outreach Committee*, `http://iscbsc.org/`.
Leading an international student organization with active chapters in more than 25 countries, organizing international research symposia, and conducting community outreach.

01/2015–present **CBioVikings – ISCB Regional Student Group Denmark**, *Co-Founder and Board member*, `http://cbiovikings.org/`.
Founding and leading the local chapter of the ISCB Student Council in Denmark, organizing community events connecting young researchers with academia and industry. E.g., an afternoon symposium on 'Tools for Reproducibility in Bioinformatics' with 60 attendees, three international speakers, and partnerships with two companies.

Honors

07/2014 and 07/2015 **Conference scholarship for Leaders of the International Society for Computational Biology (ISCB) Student Council**, *awarded by the ISCB*.
International Conference on Intelligent Systems for Molecular Biology/European Conference on Computational Biology (ISMB/ECCB) 2015 in Dublin, Ireland and ISMB 2014 in Boston, USA

07/2014 **Travel Fellowship to attend the Special Interest Group Meeting on Integrative RNA Biology**, *sponsored by the RNA Society, Boston, USA*.

03/2013–08/2013 **Scholarship for writing a Master's thesis abroad**, *awarded by the DAAD (German Academic Exchange Service)*.

07/2013 **Conference scholarship**, *awarded by the Center for Bioinformatics Saar*.
ISMB/ECCB 2013 in Berlin, Germany

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